

Plan: Extend cq-level Range from [0, 63] to [0, 81]

Context

In AOM_Q (constant quality) mode, the user-facing cq-level parameter (0-63) maps to internal qindex values (0-255) via a lookup table. Due to quality boost multipliers applied per hierarchical layer (key frame x0.25, ARF x0.40, etc.), cq-level=63 only produces qindex ~183 for key frames. The full qindex range [0, 255] is unreachable for boosted frame types.

Extending cq-level to 81 allows all frame types -- including key frames -- to reach qindex 255, giving the encoder full quantizer coverage across all hierarchical layers.

Approach: Virtual qindex > 255

Store unclamped "virtual" qindex values (259-327) in `oxcf->cq_level` for cq-levels 64-81. The rate control path computes `q_val` from the clamped qindex (255), applies the layer multiplier to get a delta, then adds the delta to the unclamped base. The final qindex is clamped to [0, 255] by existing `clamp()` calls. This produces the smooth per-layer differentiation verified in our CSV analysis.

Changes

1. Extend 'quantizer_to_qindex' table and reverse function

File: `av1/encoder/av1_quantize.c` (lines 1784-1805)

- Update comment: "0-63" to "0-81"
- Add 18 entries continuing from 255 with step +4 (unclamped virtual values):
c
259, 263, 267, 271, 275, 279, 283, 287, 291,
295, 299, 303, 307, 311, 315, 319, 323, 327,
- `av1_qindex_to_quantizer()`: change loop bound `< 64` to `< 82`, fallback return 63 to return 81

2. Clamp 'best_allowed_q' and 'worst_allowed_q'

File: `av1/av1_cx_iface.c` (lines 471-475)

`best_allowed_q` and `worst_allowed_q` feed directly into quantization and must stay in [0, 255]. `cq_level` is only used in rate control delta math and can exceed 255:

```
oxcf->best_allowed_q =  
    extra_cfg->lossless ? 0 : AOMMIN(av1_quantizer_to_qindex(cfg->rc_min_quantizer), MAXQ);  
oxcf->worst_allowed_q =  
    extra_cfg->lossless ? 0 : AOMMIN(av1_quantizer_to_qindex(cfg->rc_max_quantizer), MAXQ);  
oxcf->cq_level = av1_quantizer_to_qindex(extra_cfg->cq_level); // can be > 255
```

3. Guard 'av1_convert_qindex_to_q()' against qindex > 255

File: `av1/encoder/ratectrl.c` (line 153)

Add clamp at the top of the function so `ac_qlookup[]` is not accessed out of bounds:

```
double av1_convert_qindex_to_q(int qindex, aom_bit_depth_t bit_depth) {  
    qindex = AOMMIN(qindex, MAXQ); // clamp to valid ac_qlookup range  
    // ... rest unchanged  
}
```

4. Update range checks and defaults

File: `av1/av1_cx_iface.c`

Line	Current	New
231	'RANGE_CHECK_HI(cfg, rc_max_quantizer, 63)'	'81'
320	'RANGE_CHECK_HI(extra_cfg, cq_level, 63)'	'81'
1602	'63, // rc_max_quantizer'	'81'

5. No changes needed

- aomenc.c: CLI option definitions have no hardcoded range -- validation is in av1_cx_iface.c
- warnings.c: check_quantizer() only warns about narrow Q range
- QINDEX_RANGE, MAXQ, ac_qlookup[]: Internal qindex space [0, 255] is unchanged
- Rate control minq tables: Indexed by qindex [0, 255], unchanged
- Existing clamp() calls in rc_pick_q_and_bounds_*: Already clamp final q to [rc->best_quality, rc->worst_quality] = [0, 255]

Files to Modify

File	Changes
'av1/encoder/av1_quantize.c'	Extend table to 82 entries; update reverse lookup
'av1/av1_cx_iface.c'	Range checks 63->81; default 63->81; clamp best/worst_allowed_q
'av1/encoder/ratectrl.c'	Clamp qindex in 'av1_convert_qindex_to_q()'

Safety Analysis: Impact on Other Rate Control Modes

Mode	Uses cq_level?	Impact
Status		

AOM_CBR	No	Completely unaffected
SAFE		
AOM_VBR	No	Completely unaffected
SAFE		
AOM_CQ	Yes (comparisons)	cq_level used in 'if (q < cq_level)' -- final q clamped to [0, 255]
SAFE		
AOM_Q (1-pass)	Yes (direct)	Passes cq_level to 'av1_convert_qindex_to_q()' -- requires Fix #3
SAFE with fix		
AOM_Q (2-pass)	Yes (direct)	Same as 1-pass AOM_Q
SAFE with fix		

Key safety guarantees:

- av1_ac_quant() in quant_common.c already has internal clamp(qindex + delta, 0, MAXQ) -- defense-in-depth
- All rc_pick_q_and_bounds_* functions clamp final q to [rc->best_quality, rc->worst_quality] = [0, 255]
- best_allowed_q/worst_allowed_q flow into rc->worst_quality which is used as array index for inter_minq[]/rtc_minq[] (256 elements) -- must stay <= 255 (Fix #2)
- cq_level is never used as a direct array index -- only in arithmetic and as input to av1_convert_qindex_to_q() (Fix #3)

Verification

1. Build: cmake path/to/aom && make -j\$(nproc)
2. Original range: ./aomenc --end-usage=q --cq-level=32 -o out.ivf input.y4m -- no regression
3. Extended range: ./aomenc --end-usage=q --cq-level=70 -o out.ivf input.y4m -- runs, produces valid output
4. Max cq: ./aomenc --end-usage=q --cq-level=81 -o out.ivf input.y4m -- key frames reach qindex ~255
5. Out of range: ./aomenc --end-usage=q --cq-level=82 -- rejected with error
6. CBR mode: ./aomenc --end-usage=cbr --target-bitrate=1000 -o out.ivf input.y4m -- no regression
7. CQ mode: ./aomenc --end-usage=cq --cq-level=70 -o out.ivf input.y4m -- runs correctly
8. Unit tests: make runtests